

1. Using the Euclidean algorithm, show that $\gcd(48, 18) = 6$. Using this, find $x, y \in \mathbb{Z}$ so that $6 = 48x + 18y$.
2. Using Bézout's theorem, show if $\gcd(a, m) = 1$ then there exists $x \in \mathbb{Z}$ so that $ax \equiv 1 \pmod{m}$.
3. Show that if $n \mid ab$ and $\gcd(n, a) = 1$, namely n and a are coprime, then $n \mid b$.

4. Show that if $ac \equiv bc \pmod{m}$ then:

$$a \equiv b \pmod{\left(\frac{m}{\gcd(m, c)}\right)}$$

5. Suppose n, m have the following prime factorizations:

$$n = \prod_{i=1}^k p_i^{a_i}, \quad m = \prod_{i=1}^k p_i^{b_i},$$

Using this, find a formula for $\gcd(n, m)$ and $\text{lcm}(n, m)$ in terms of these prime factors. With this, show $\gcd(n, m) \cdot \text{lcm}(n, m) = nm$.

6. (Fun, unrelated, and challenging). Suppose we have a circular race track with circumference one. Along the track are n gas cans at angles $x_1, \dots, x_n \in [0, 2\pi)$. The can at location x_i contains enough fuel to travel distance $d_i \geq 0$. There is exactly enough gas across the n cans to traverse the track: $\sum_{i=1}^n d_i = 1$. We aim to drive a car clockwise along this track starting with no fuel and collecting gas from the gas cans along the way. Prove or disprove: for any choice of $n, \{x_i\}_{i=1}^n, \{d_i\}_{i=1}^n$ there exists a starting point $z \in [0, 2\pi)$ so that our car can make it all the way around the track without running out of fuel.

